

UbiComp4VRU: Ubiquitous Computing for Vulnerable Road Users

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Abstract

Achieving sustainable urban transportation requires not only urban planning but also technological innovation, particularly for vulnerable road users (VRUs) such as cyclists, pedestrians, and e-scooter riders. While active mobility (i.e., cycling and walking) offers environmental and health benefits, VRUs face a higher risk of serious and fatal accidents than vehicles. This full-day workshop will examine how ubiquitous computing, sensing, and communication technologies can improve VRU safety, promote active mobility, and support the development of a sustainable and resilient transportation system while addressing socio-technical aspects such as legal, ethical, psychological, and normative considerations. Therefore, we will invite two keynote speakers, one from technical and one from socio-technical science, and prepare an open discussion session to foster interdisciplinary exchange. The discussion will encourage participants to explore technical innovations and socio-technical implications for enhancing VRU safety and promoting sustainable mobility.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; • **Computing methodologies** → Artificial intelligence; Distributed artificial intelligence; • **Hardware**; • **Social and professional topics** → User characteristics;

Keywords

VRU, Ubiquitous Computing, Human Activity Recognition, Sensing, Communication

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1 Workshop Title and Acronym

1st International Workshop on Ubiquitous Computing for Vulnerable Road Users (UbiComp4VRU) Workshop at UbiComp/ISWC 2025

2 Brief Description of Technical Issues

Active mobility, such as cycling and walking, is an important part of a sustainable transportation system in the city of tomorrow. This offers several benefits, such as improving air quality and promoting health through increased physical activity, among others. However, traffic accidents, especially those involving vehicles, are the number one cause of death worldwide for children and young adults up to 29 years of age [1]. Road safety is an essential key to successfully promoting active mobility. In recent years, the proportion of seriously injured or killed occupants in cars has been significantly reduced by AI-based (AI: artificial intelligence) assistance systems, such as emergency brake assistants.

However, cyclists, pedestrians, and e-scooter drivers (i.e., vulnerable road users (VRU)) remain highly vulnerable without protective systems. Enhancing their safety requires developing innovative solutions, including ubiquitous, pervasive, and wearable computing, implementing appropriate infrastructure measures, and targeted training programs for cyclists, pedestrians, and e-scooter drivers. Furthermore, improving the detection and anticipatory modeling of cyclist, e-scooter driver, and pedestrian behavior in AI-driven autonomous systems can contribute to risk mitigation. Beyond safety measures, fostering a shift from car usage to active mobility requires strategies to enhance motivation for cycling and walking. Advancing novel technologies, applications, and wearable devices

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can play a crucial role in increasing road safety, promoting sustainable mobility, and supporting the development of more resilient urban environments.

While there have been contributions targeting some of these issues, many challenges remain unsolved. The proposed workshop will explore the challenges of:

- The design, development, and deployment of ubiquitous, pervasive, and wearable computing technologies that contribute to a safer and more sustainable urban environment
- Communication and sensing technology for ubiquitous computing approaches to improve VRU safety
- The development and embedding of new sensors and devices for VRU's
- New AI algorithms and approaches for VRUs' activity detection
- Innovative technology, such as applications, sensors, and wearable devices motivating to cycle and walk
- Consideration of VRU safety in scenarios on edge cases, like around footpaths, parking areas, or dooring
- Interdisciplinary approaches toward traffic safety and creating a sustainable transportation system (e.g., legal, ethical, psychological, normative, etc.)

We aim to have unique contributions addressing these research topics and provide a discussion space to facilitate collaboration among researchers interested in this field.

Ubiquitous computing for VRU is a multidisciplinary area and is greatly interesting to various research fields. This important topic has been discussed in some conferences, e.g., *ACM Conference on International Conference on Mobile and Ubiquitous Multimedia*, *ACM Conference on Conference on Automotive User Interfaces and Interactive Vehicular Applications*, *IEEE International Conference on Pervasive Computing and Communications*. Recently, workshops in the field of computer science have discussed the development of smart mobility within the Workshop *International Workshop on Pervasive Computing for Vehicular Systems (PerVehicle)* – held in conjunction with the *IEEE International Conference on Pervasive Computing and Communications* conference. Another example is the *ACM SIGSPATIAL International Workshop on Sustainable Urban Mobility*.

Last year, there was no workshop at UbiComp/ISWC dealing with ubiquitous computing for VRUs. While the named conferences and workshops deal primarily with the technical details, UbiComp/ISWC offers an interdisciplinary, innovative, and creative environment for the workshops' topic. The proposed workshop matches many key topics of UbiComp/ISWC, such as the development of new wearables, ubiquitous and pervasive computing, as well as the understanding of human experiences and social impacts.

Thus, this workshop offers an interdisciplinary discussion panel for researchers from various fields in computer science, urban planning, psychology, and sustainability. Using this space, they will have a chance to meet and discuss the state-of-the-art together. Researchers, for example, from psychology, can provide valuable information about human behavior in traffic situations, which may help researchers from the field of computer science to understand and predict human behavior better. This interdisciplinary cooperation will further improve the development of ubiquitous wearables

and applications for the sustainable transportation system in the city of tomorrow.

3 Submissions and Program Committee Members

Submitted papers will be evaluated with the help of the proposed TPC. The evaluation will be based on the guidelines of UbiComp and ISWC. Accordingly, the papers will be evaluated, for example, for their (technical) correctness, originality, and innovation. Furthermore, their thematic relevance to the workshop and the conference, and their quality will be examined.

3.1 Technical Program Committee

Sian Lun Lau

School of Engineering and Technology
Sunway University

Dr. Yanqiu Huang

Pervasive Systems Group
University of Twente

Zengyi Han

Dalian Maritime University

Dr. Chelsea Dobbins

School of Information Technology and Electrical Engineering
The University of Queensland

Prof. Stephan Sigg

School of Electrical Engineering
Aalto University, Finland

Prof. Kristof van Laerhoven

Department of Electrical Engineering and Computer Science
University of Siegen

M.Sc. Marie-Christin Oczko

Chair for Telecommunication Networks
Technical University Berlin

4 Communities and Groups attracted

DyNaMo Workshop focuses on the novel research of active mobility, and is aimed at attracting researchers from the fields of machine learning, wearable development, bicycle safety, autonomous driving, urban planning, psychology, and sustainability, who will deal with the technical design and the effects of technology on pedestrian, bicycle, and e-scooter driver safety with the joint goal of creating a sustainable transportation system in the city of tomorrow. This workshop is of great interest to the scientific community in the areas of computer science, artificial intelligence, human-computer interaction, pervasive and ubiquitous computing, where active mobility is considered an important/necessary context for context-aware applications. Further, as normative implications are also within the focus, social scientists and scientists who research within the legal norm are also welcome to discuss and contribute to this workshop.

5 Planned Format and Strategy of the Workshop

The open workshop is proposed to be a one-day discussion, including presentations from two invited keynote speakers and authors of the accepted papers (see Table 1). We also offer an open discussion session about new approaches, challenges, and future cooperation.

Time	Session	Note
09:00 – 09:15	Opening	
09:15 – 10:30	Keynote speaker and discussion	Note: The keynote speaker is an expert in the field of ubiquitous computing and is working on bicycle safety and sustainability. Potential Keynote Speaker: to be announced
10:30 – 11:00	Coffee Break	
11:00 – 12:30	Six presentations from authors of accepted papers, 10-minute presentation, and a 5-minute discussion	Note: the morning discussion is more focused on technical aspects like applications/approaches/models/algorithms, new devices and wearables for active mobility and sustainable transportation systems
12:30 – 14:00	Lunch	
14:00–15:00	Keynote speaker and discussion	Note: the keynote speaker is an expert in the field of transport economics and psychology. This keynote works on (inclusive) infrastructure, transport development (innovation and safety), and human factors. Potential Keynote Speaker: to be announced
15:00 – 15:30	Two presentations from authors of accepted papers, 10-minute presentation, and a 5-minute discussion	Note: the afternoon discussion is more focused on normative aspects, the human experiences, and the social impacts of ubiquitous, pervasive and wearable computing technologies for sustainable transportation systems
15:30 – 16:00	Coffee Break	
16:00 – 17:30	Open discussion	

Table 1: Conference schedule based on the breaks of the last conference

As the research topic is currently of high interest, we expect approximately 40 attendees.

The morning program begins with a presentation by an invited keynote speaker, an expert in ubiquitous computing who works on bicycle safety and sustainability. The morning presentation and discussion focus more on the technical aspects like applications, approaches, models, algorithms, new devices, and wearables for active mobility and sustainable transportation systems. The afternoon program begins with a presentation from an invited keynote speaker, an expert in transport economics and psychology. This keynote works on (inclusive) infrastructure, transport development (innovation and safety), and human factors. The afternoon presentation and discussion focus more on the normative aspects, the human experiences, and the social impacts of ubiquitous, pervasive, and wearable computing technologies for sustainable transportation systems. We close the workshop with an open discussion, inviting all attendees to participate. Therefore, we will prepare some research questions to be discussed, as well as take up and discuss in more depth some exciting discussion questions that have arisen during the workshop. We will make the results of our joint discussion available to participants after the workshop via our workshop's website.

6 Publicity Plan

- Call for paper: April 27, 2025
- Paper submission deadline: June 15, 2025
- Review deadline for TPC: June 24, 2025
- Paper notification: June 29, 2025
- Camera Ready Deadline: July 30, 2025
- Workshop: October 12-13, 2025

We will distribute the Call for Paper through mailing lists like *mycolleagues* and *announcements* and through our network, the TPCs, social media, such as LinkedIn, and other platforms.

7 Organizers Biography



Dr. Judith Simone Heinisch. is a Post-Doc at the Department for Communication Technology at the University of Kassel, Germany. She received her B. Sc. and M. Sc. in Computer Science, focusing on Activity Recognition and Data Mining, from the University of Kassel in 2014 and 2017. In 2023, she finished her doctorate in Computer Science on the topic *Algorithms for Emotion Recognition*. Her research areas of interest include Ubiquitous and Pervasive Computing, Machine Learning, Data Mining, Emotion Recognition, and Activity Recognition. Currently, her research topic is on vulnerable road user safety and human activity recognition.



Prof. Dr. Klaus David. (born 1961 in Frankfurt am Main, Germany) was, after studying Physics at the University of Siegen, a researcher at IMEC, in Ghent (Belgium) for four years and received his doctorate in Engineering at the University of Siegen. In 1988, after six years as a project and team leader at T-Mobile, he was appointed to the Chair of Mobile Systems at the Technical University of Brandenburg. It was here that he initiated the still successful lesswire AG and was a Department Head at IHP. In total, he has spent about five years working abroad (UK, Belgium, USA, and Japan). Since

May 2000, Prof. Dr. David has directed the Chair of Communication Technology (ComTec) at Kassel University. He has registered over ten patents, written two books and published more than 200 scientific papers. He is an active member of many international organizations such as the Institute of Electrical and Electronics Engineers (IEEE, BoG 2015-2017, and EIC 2015 - 2018), Wireless World Research Forum (WWRF). Prof. Dr. David is also a reviewer of the German Research Society (DFG), the German Ministry of Education and Research (BMBF) and the EU. His research includes Mobile Computing, Pervasive Computing, and Context Awareness. In 2021, he was the General Chair of the IEEE PerCom.



Dr. Si Zuo. is a Post-Doc at the Department of Information and Communications Engineering at Aalto University, Finland. She received her M. Sc in Communication Engineering from Aalto University in 2018. She obtained her PhD from Aalto University in 2024 on the topic of balancing the privacy and utility of smart devices. Her work focuses on the intersection

of wearable sensing, human activity recognition, generative models for multimodal time-series data, and privacy protection in IoT devices.



Prof. Dr. Flora Salim. is professor at the CISCO Chair in Digital Transport, School of Computer Science and Engineering, UNSW, and Honorary Professor, School of Computing Technologies, RMIT University. She is a member of the Australian Research Council (ARC)

College of Experts. Her research is in the intersection of human-centred computing and machine learning for behaviour intelligence with multimodal sensor data, with applications in intelligent assistants, health monitoring, smart cities, smart buildings, spatio-temporal recommendation systems, and AI on the edge. She was a Humboldt-Bayer Fellow, Humboldt Fellow, Victoria Fellow, and ARC APDI Fellow. She obtained her PhD from Monash University in 2009.



Hiruni Kegalle. is a PhD candidate at the School of Computing Technologies, RMIT University, Melbourne. She received her B. Sc. in Computer Science from the University of Colombo School of Computing, Sri Lanka in 2019. Her research areas of interest include human-computer interaction, machine learning, multimodal behavioral data analysis, and spatio-temporal data analysis.

8 Website Address

<https://www.comtec.eecs.uni-kassel.de/ubicomp4vru-workshop/>

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